

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

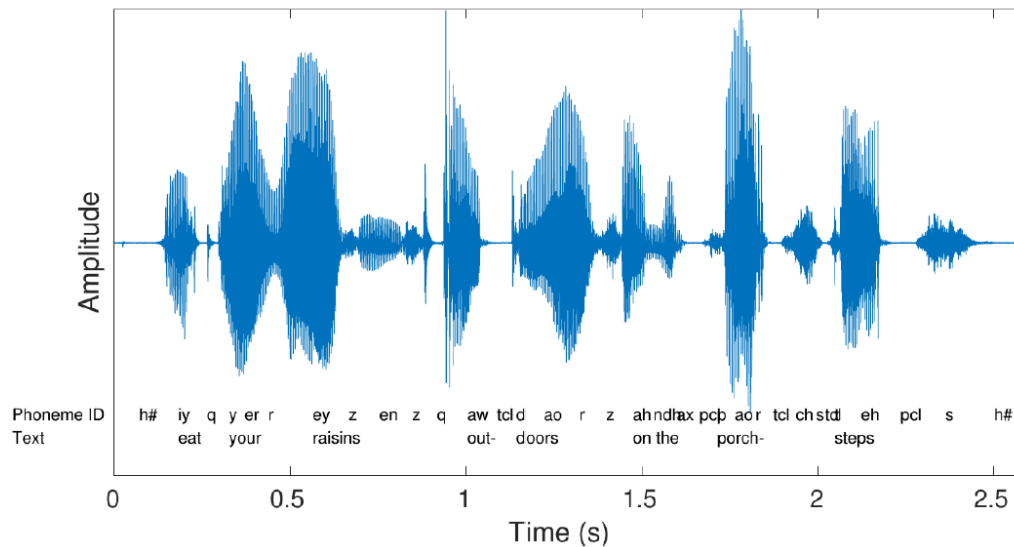
HIDDEN MARKOV MODELS

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Sequence Data

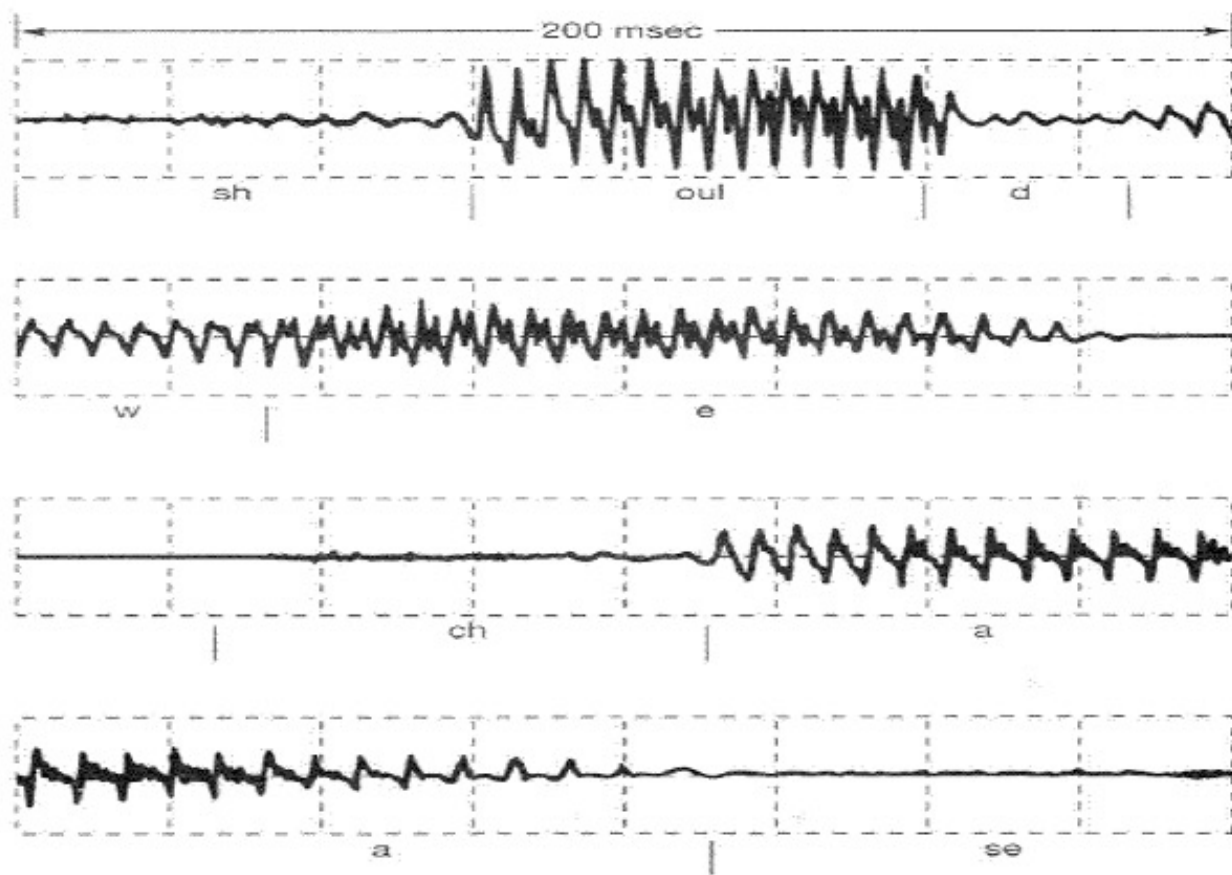


Phonemes are the basic units of speech, e.g.

$\text{TOMATO} = [\text{T}] + [\text{OW}] + [\text{M}] + [\text{AA}] + [\text{T}] + [\text{OW}]$

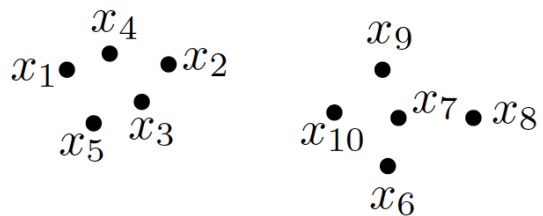
There are 40-60 phonemes.

Speech Representation



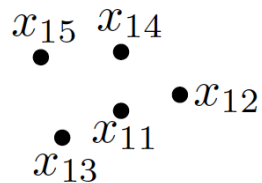
Temporal Structure in Sequences

Each data point is a sequence $x = (x_1, x_2, \dots)$ of vectors.



S_1

S_2



S_3

Hidden Markov Models: Definition

A **hidden Markov model** (HMM) is specified by:

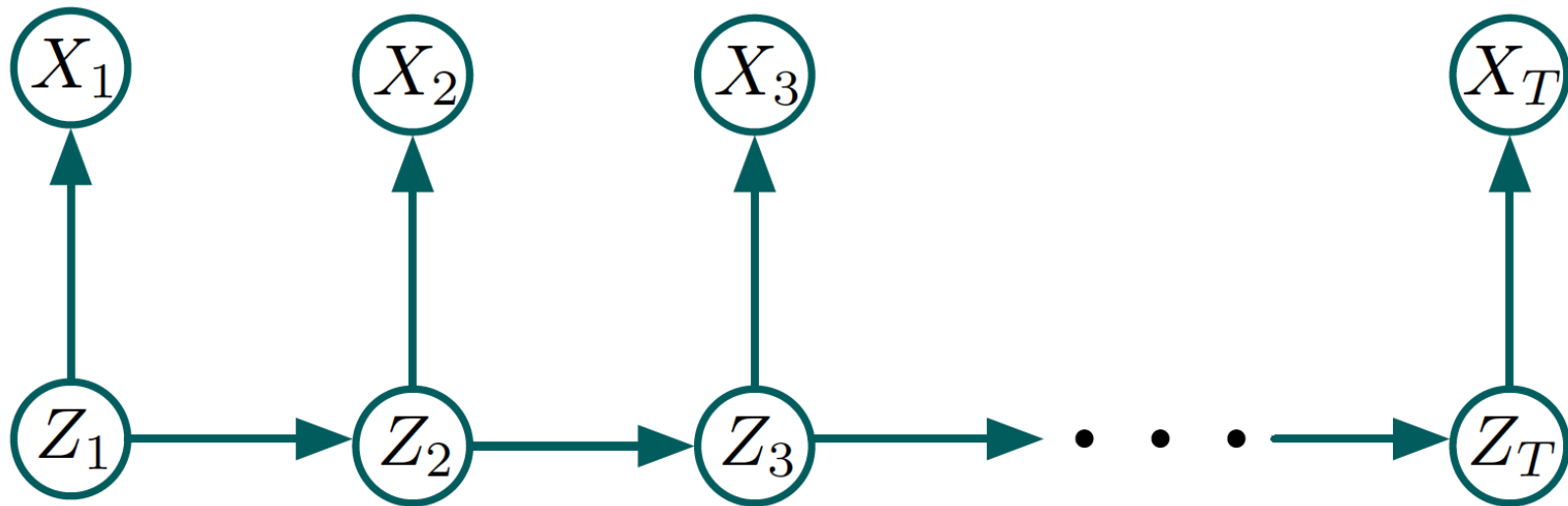
- A set of (say) k states
- Distributions $P_1, \dots, P_k$ associated with each of the states
- An initial distribution over states, $\pi \in \Delta_k$
- Transition probabilities given by a $k \times k$ matrix:

$$A_{ij} = \Pr(\text{transition from state } i \text{ to state } j)$$

This HMM generates a sequence $X = (X_1, X_2, \dots)$ as follows:

- Pick initial state $Z_1 \sim \pi$
- For $t = 1, 2, \dots$:
 - $X_t \sim P_{Z_t}$
 - Choose next state Z_{t+1} according to A , i.e, $\Pr(Z_{t+1} = j | Z_t = i) = A_{ij}$

Latent Variable Model



Inference

Given an HMM, we can easily sample sequences from it.

But we also want to answer queries:

- What is the probability of a particular sequence, $P(X_1, \dots, X_T)$?
- What is the distribution over hidden states at time t , given a sequence, $P(Z_t | X_1, \dots, X_T)$?
- What is the most likely sequence of hidden states for a given sequence, $\arg \max P(Z_1, \dots, Z_T | X_1, \dots, X_T)$?

All can be answered efficiently because the model is tree-structured.

Learning

Given a collection of sequences (X_t) , find maximum-likelihood parameters $\pi, A, \{P_j\}$.

- This is NP-hard.
- There is an EM algorithm for it.

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Outline of EM for HMMs:

- At any given time, maintain an estimate of the parameters.
- Based on the current estimate, use efficient inference to compute statistics:
 - How much time is spent in each state?
 - What is the distribution in each state?
 - What is the starting distribution over states?
 - How often does each possible inter-state transition occur?
- Use this information to update the parameters, and repeat until convergence.